IndiaWorks Unified SaaS - Functional Requirements Document (FRD)

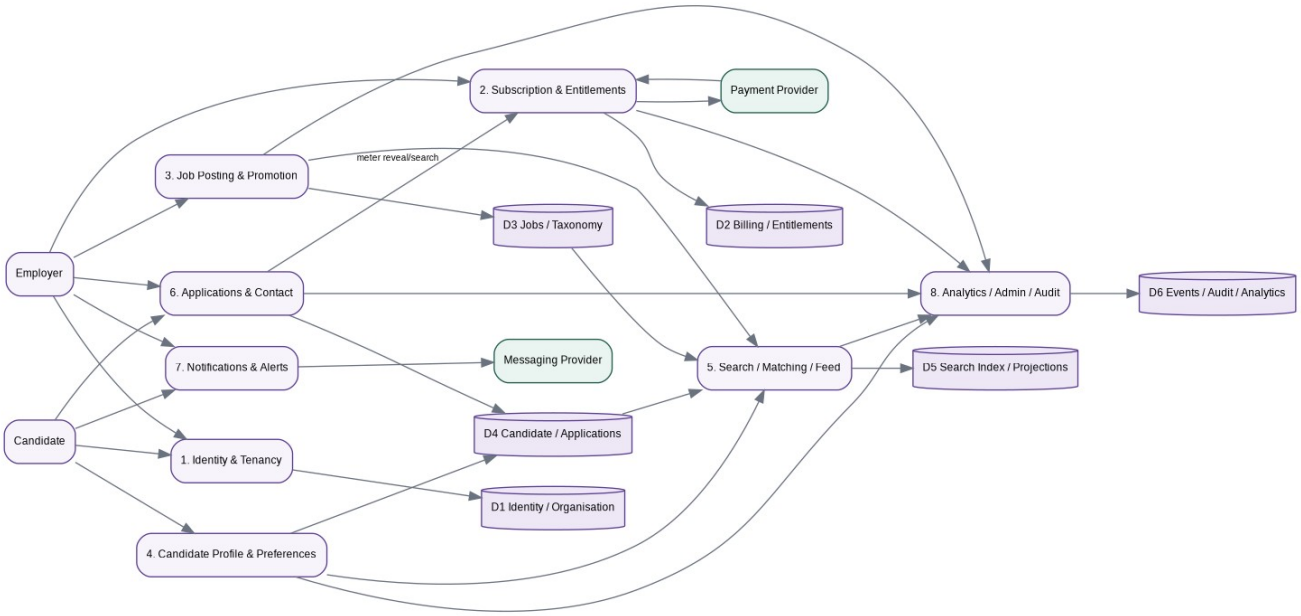
Product flows, rules, states and acceptance criteria

IndiaWorks | Unified SaaS Architecture v7
Generated 25 September 2026
Baseline: v5 Unified SaaS + v6 Product/Technical Review + v7 Five-Candidate Commercial Gate

1. Purpose

This FRD turns the unified IndiaWorks business model into implementable functional behavior across candidate, employer, subscription, job, matching, promotion, application and admin domains.

2. Functional Architecture



3. Identity and Session Management

ID	Function	Functional behavior	Key validation / acceptance
FR-AUTH-01	Candidate OTP login	Request OTP to mobile; verify challenge; create/login UserAccount + CandidateProfile; issue access/refresh session; route to jobs.	OTP expiry/attempt limits; idempotent candidate creation; verified mobile unique.
FR-AUTH-02	Employer OTP login	Support verified mobile and/or email OTP.	Identifier uniqueness; employer may hold multiple organisation memberships.
FR-AUTH-03	Session refresh/logout	Rotate/refresh tokens and revoke on logout/security event.	Revoked refresh token cannot mint new sessions.
FR-AUTH-04	Role context	Every employer operation resolves organisation membership and permission.	No organisation ID trust from client without membership authorization.

4. Organisation and Recruiter Management

ID	Function	Functional behavior	Acceptance
FR-ORG-01	Create organisation	Owner creates company with name, size band, sector, industry and locations.	Organisation created once; owner membership added transactionally.
FR-ORG-02	Verification	Store PAN/GSTIN/CIN/LLPIN evidence and verification state separately from public company fields.	Sensitive identifier encrypted/hashed; masked display only.
FR-ORG-03	Membership/RBAC	OWNER, ADMIN, RECRUITER, VIEWER, BILLING_ADMIN.	Resolvers enforce both membership status and action permission.
FR-ORG-04	Seats	Active recruiter/member seats consume seat entitlement where configured.	Invite blocked or upgrade prompted when hard seat limit reached.

5. Subscription, Trial, Billing and Entitlements

ID	Function	Functional behavior
FR-SUB-01	Plan catalogue	Expose active plan versions with interval, public features and price metadata.
FR-SUB-02	Trial	Provision an eligible organisation free-evaluation/trial state. Candidate profile access includes exactly five lifetime distinct free unlocks with reset NONE; sixth distinct candidate requires paid entitlement.
FR-SUB-03	Checkout	Create provider checkout/order for selected monthly/yearly plan or add-on; never accept client-supplied paid status.
FR-SUB-04	Webhook reconciliation	Persist provider event in webhook inbox; process idempotently; update payment transaction/invoice/subscription.
FR-SUB-05	Entitlement check	Central service returns ALLOW / DENY / ALLOW_AND_METER with remaining quota and reset time.
FR-SUB-06	Usage meter	Write immutable EntitlementUsageEvent with

ID	Function	Functional behavior
		idempotency key for search/reveal/promotion/export units.
FR-SUB-07	Upgrade/downgrade	Apply configured immediate or next-renewal policy; preserve audit trail.
FR-SUB-08	Cancellation	Support cancel-at-period-end and administrative suspension.
FR-SUB-09	Past due	Grace policy configurable; prevent new premium actions after grace while preserving historical data.
FR-SUB-10	Billing dashboard	Show current plan, interval, period/trial end, usage/limits, invoices and change-plan actions.

6. Job Posting and Publishing

ID	Function	Functional behavior	Important rule
FR-JOB-01	Draft job	Build job from canonical Role -> Specialisation -> WorkContext plus independent work/engagement/location/pay requirements.	Collar is optional market tag only.
FR-JOB-02	Dynamic template	Role selection loads recommended skills, credentials, assets and candidate questions.	Employer may edit preference levels within policy.
FR-JOB-03	Validation	Validate vacancy count, compensation, locations, required credentials, contact windows and description.	Return field-level validation issues.
FR-JOB-04	Publish	Check organisation, permission, subscription status and active-job entitlement.	Publish and entitlement meter occur in one application transaction boundary.
FR-JOB-05	Lifecycle	DRAFT -> PUBLISHED -> PAUSED/CLOSED/FILLED/EXPIRED.	Every change recorded in status history.

7. Job Promotion

ID	Function	Functional behavior	Acceptance
FR-PRO-01	Promote job	Create timed promotion against a published eligible job using included credit or paid add-on.	Promotion cannot extend beyond closed job; entitlement/credit consumed idempotently.
FR-PRO-02	Promotion placement	Support BOOSTED_SEARCH, FEATURED_FEED, URGENT_BADGE or future configured types.	Placement products are configuration, not columns on JobPosting.
FR-PRO-03	Ranking	Promotion boost applied after hard eligibility and baseline relevance.	Promotion cannot surface an otherwise ineligible job/candidate pair.
FR-PRO-04	Reporting	Track promotion impressions, qualified views, applications, contacts and spend/credits.	Metrics attributable to promotion id.

8. Candidate Onboarding and Profile

ID	Function	Functional behavior	Acceptance
FR-CAN-01	Progressive onboarding	OTP opens feed immediately; profile may be sparse.	No CV/Aadhaar/PAN/full education gate.
FR-CAN-02	Occupation	Support multiple CandidateOccupation records with one primary.	Occupation separate from skills.
FR-CAN-03	Preferences	Capture role interests, location/radius, employment types, work modes, shifts, expected pay, availability/notice.	All optional until needed by a job/apply path.
FR-CAN-04	Discoverability	Default APPLIED_JOBS_ONLY; candidate may opt into SEARCHABLE_VERIFIED_EMPLOYERS.	Separate talent-pool consent.
FR-CAN-05	Credentials/assets	Credential and asset records remain separate; collect just-in-time.	Driving licence credential; bike/phone actual asset.

9. Candidate Job Discovery

ID	Function	Functional behavior
FR-FEED-01	Feed	Return cursor-paginated jobs from search projection ranked by eligibility/relevance, freshness and bounded promotion.
FR-FEED-02	Filters	Role/family, qualification/course, experience, engagement, work mode, location/radius, salary, shift, urgency, employer sector/context.
FR-FEED-03	Job detail	Return structured description/pay/location/requirements/contact policy and candidate-specific match result.
FR-FEED-04	Save/share	Persist save; record share channel asynchronously where appropriate.
FR-FEED-05	Alerts	Persist saved filter + explicit notification-channel opt-in + frequency.

10. Application and Contact

ID	Function	Functional behavior	Acceptance
FR-APP-01	Prepare apply	Evaluate HARD/WARN/MISSING requirements before mutation.	No side effect; clear reasons returned.
FR-APP-02	Apply	Create one application per candidate/job with idempotency key, apply-time snapshots and APPLIED event.	Duplicate retries return same application.
FR-APP-03	Soft mismatch	Education/experience/language/distance preferences default to warning when policy allows.	Candidate can continue.
FR-APP-04	Hard block	Explicit mandatory legal/safety/credential rule may block.	Candidate sees exact action needed.
FR-APP-05	Status	Employer transitions application through configured state machine.	Every state change creates event.
FR-APP-06	Contact employer	Candidate call/message respects job contact	Contact event written.

ID	Function	Functional behavior	Acceptance
		policy/window.	
FR-APP-07	Withdraw	Candidate may withdraw non-terminal application.	WITHDRAWN event + timestamp.

11. Employer Candidate Discovery

ID	Function	Functional behavior	Acceptance
FR-TAL-01	Search	Search structured candidate index by job or explicit filters.	Requires candidate-search entitlement and searchable consent where applicable.
FR-TAL-02	Job-based matches	Use job requirements to generate ranked candidate list.	Hard eligibility first, then structured relevance/ranking.
FR-TAL-03	Masked card	Return role, skills, experience, approximate location, availability, verification indicators without direct contact PII.	PII not in search index.
FR-TAL-04	Profile view	Record auditable profile view and enforce visibility.	Subscription access checked.
FR-TAL-05	Contact reveal	Reveal permitted contact after consent/visibility + entitlement check; meter once.	Idempotent reveal and CandidateAccessAudit.
FR-TAL-06	Shortlist	Employer can save/shortlist candidate against organisation/job.	Does not itself reveal contact.

12. Matching and Ranking Rules

```
candidate_job_rank =
  eligibility_gate(
    credential, legal/safety requirement, job state, location policy
  )
  -> structured_relevance(
    role/specialisation, skills, experience, language,
    availability, compensation, distance
  )
  -> quality_adjustments(
    verification, profile freshness, job freshness
  )
  -> bounded_commercial_boost(
```

```
    active promotion only
)
```

The ranking service must return explanation codes so clients can distinguish eligibility, relevance and promotional placement. A promotion flag is always disclosed in the result metadata for analytics/UI.

13. Admin and Support

- Plan/version management and entitlement matrix.
- Promotion product configuration.
- Taxonomy version/alias management.
- Organisation and job moderation/suspension.
- Candidate privacy/support workflows and visibility overrides only under auditable policy.
- Payment/subscription reconciliation dashboard.
- Usage and access audit search.
- Feature flags and policy configuration.

14. State Machines

Domain	States / transitions
Subscription	TRIALING -> ACTIVE -> PAST_DUE -> ACTIVE or SUSPENDED; ACTIVE -> CANCEL_AT_PERIOD_END -> CANCELLED; TRIALING -> EXPIRED.
Job	DRAFT -> PUBLISHED -> PAUSED -> PUBLISHED; PUBLISHED -> FILLED/CLOSED/EXPIRED.
Promotion	SCHEDULED -> ACTIVE -> COMPLETED; SCHEDULED/ACTIVE -> CANCELLED; ACTIVE -> PAUSED where supported.
Application	APPLIED -> VIEWED -> SHORTLISTED -> CONTACTED -> INTERVIEW_SCHEDULED -> OFFERED -> HIRED; REJECTED/WITHDRAWN/EXPIRED terminal alternatives.

15. Functional Acceptance - MVP

1. Employer can OTP-login, create organisation, activate a trial, publish a structured job, and see entitlement usage.
2. Candidate can OTP-login and see a relevant job feed before completing a full profile.
3. Candidate can apply with soft warnings and only just-in-time mandatory data collection.
4. Employer can see applications and subscription-gated matching candidates.
5. Candidate cards are masked until a permitted, metered reveal/contact action.
6. A promoted job gains a bounded ranking/placement boost only when otherwise eligible/relevant.
7. Payment webhook retries do not double-activate plans or double-meter usage.
8. All candidate access/reveals and subscription changes are auditable.

v7 commercial gate - five free candidates

Rule: A verified organisation/legal entity receives five lifetime free DISTINCT candidate unlocks. The sixth distinct candidate requires a paid monthly/yearly/add-on CANDIDATE_UNLOCK entitlement.

Rule	v7 behavior
Masked discovery	Applicant/search result cards and candidate counts are visible in masked form without consuming an unlock.
Consumption key	Only first full-profile access of a distinct (OrganisationId, CandidateProfileId) consumes candidate access.
Scope	The five free candidates are shared by every recruiter, every job and every candidate-access surface in the organisation.
No reset	Free allowance never resets by month, billing period, new recruiter, new job, device, cancellation or re-sign-up.
Duplicate candidate	Re-opening the same candidate or finding the same candidate against another job does not consume again.
Privacy first	Consent/application context, discoverability and candidate-organisation block rules are evaluated before payment/entitlement.
Paid access	Paid plan/add-on entitlements can grant further CANDIDATE_UNLOCK units or unlimited access.
Contact	Candidate contact reveal remains separately consent checked, entitlement checked and audited.
Abuse control	Commercial identity fingerprint/risk controls prevent recreating organisations purely to regain five free candidates.

17. Candidate access and paywall functional requirements

ID	Function	Functional behavior	Acceptance
FR-PAY-10	Candidate access status	Return freeLimit=5, freeUsed, freeRemaining, paidAccessActive and paywallActive for an organisation.	All recruiter users observe one shared organisation counter.
FR-PAY-11	Masked list	Applicant/candidate-search list returns masked cards and access state.	Scrolling/opening masked card does not create candidate unlock usage.
FR-PAY-12	Unlock	unlockCandidateProfile atomically evaluates privacy, existing unlock, five-free allowance and paid entitlement.	First five distinct candidates FREE_UNLOCKED; sixth PAYWALL_REQUIRED without paid plan.
FR-PAY-13	Idempotency	Organisation+Candidate is the consumption key; idempotency key handles request retry.	Concurrent duplicate requests create one unlock record maximum.
FR-PAY-14	Paid retry	After provider webhook activates plan, employer can retry protected action.	Retry yields PAID_UNLOCKED and one usage event.
FR-PAY-15	Contact reveal	Contact reveal requires candidate unlock, privacy and contact	Reveal audit exists; no contact PII in search index or analytics event.

		entitlement/policy.	
FR-PAY-16	Abuse prevention	Free allowance is legal-entity scoped and cannot be reset through new recruiter/job/device/account.	Duplicate verified commercial identity is held/reused/flagged according to policy.

18. v7 job discovery and filter requirements

Dimension	Behavior
Taxonomy	Family/Role/Specialisation/WorkContext are separate selectable dimensions backed by canonical taxonomy and aliases.
Geography	City/locality/PIN/radius; distance calculation and deterministic geographic filtering.
Work	WorkMode and EngagementType are independent.
Experience	Any/Fresher/Experienced UX maps to numeric min/max months and employer preference/gate.
Education	Education level and course/field (including ITI/CA/MBA/BTech etc.) are separate.
Compensation	Filter on normalized monthly equivalent plus original pay frequency/bonus/incentive.
Schedule	Day/Night/Rotational/Flexible and timing/weekly off.
Freshness/Urgency	Date posted/refresh and hiring urgency are independent.
Capabilities	Languages, skills, credentials and assets are structured.
Employer	Verified/direct source, employer sector, industry and company-size facets.
Benefits	PF/ESI/insurance/food/accommodation/transport etc. are structured.
Facets	jobSearchFacets returns counts contextual to the active filter/query/geography.

19. Additional integrated v6 flows

- Persistent session/device management, session rotation/revocation and candidate/employer persona switch.
- Organisation invitation lifecycle with expiry, role and seat-entitlement checks.
- Stable JobPublicLink/JobShareLink with mobile app deep-link and web fallback.
- SavedJobSearch -> JobAlertSubscription -> JobAlertRun -> JobAlertDelivery watermark/deduplication pipeline.
- EmployerCandidateRelationship with owner, stage, notes, tags and links to multiple jobs/applications.
- Interview and Offer entities for hiring lifecycle rather than status-only records.
- MarketplaceReport and CandidateOrganisationBlock enforced across search, alerts and contact.

20. Updated MVP acceptance / boundary suite

Scenario	Expected result
Unlock #1	#1 FREE_UNLOCKED; freeRemaining=4.
Unlock #5	#5 FREE_UNLOCKED; freeRemaining=0.
Open #5 again	ALREADY_UNLOCKED; counter remains 5.

Same #5 from another recruiter/job	ALREADY_UNLOCKED; counter remains 5.
Unlock #6 unpaid	PAYWALL_REQUIRED; no full profile returned.
Activate paid plan then retry #6	PAID_UNLOCKED according to paid entitlement.
Candidate blocks organisation	BLOCKED_BY_POLICY before entitlement evaluation.
Job filter changes	Facet counts recompute for active context; cursor order remains deterministic.
Duplicate apply/unlock request	Idempotent result; no duplicate application/unlock/usage event.